ASSESSMENT A1 · STUDENT PACKAGE

Specification Portfolio

Week 3 · Individual assessment

SUBMISSION	WEIGHT	FORMAT
PDF via LMS, end of Week 3	10% of Semester 1 grade	Single PDF: all 5 specs + flowchart

What this assessment is testing

A1 is the first thing you submit in this programme. It does not ask you to write code, call an API, or build anything. It asks you to do something harder: say precisely what an AI system should do before anyone builds it.

It tests two things.

1. Can you write a specification that someone else could build from?

A specification is a contract. It describes inputs, outputs, constraints and edge cases so precisely that a developer reading it knows exactly what to build, and a tester knows exactly how to verify it. A spec that says *'return a helpful response'* is not a specification. A spec that says *'return two sentences, maximum 40 words; the first states the strongest positive, the second states the strongest negative'* is.

2. Can you think through what happens when inputs go wrong?

Every AI system will receive inputs it was not designed for. An edge case is an input at the boundary of what the system handles, and it must name the exact trigger and the exact return message. *'Handle invalid input'* is not an edge case. *'If the input is fewer than 20 words → return: Review is too short to summarise.'* is.

The task

Write one task specification for each of these five domains: **Health, Transport, Education, Food and Scheduling.** Every specification must have exactly five fields:

Field	What it means
Task	One sentence: what does the AI do?
Input	What does it receive? Format, language, length. Be specific.
Expected output	What does a correct response look like? Specific enough to test.
Constraints	What must it not do? State these as specific prohibitions.
Edge cases	**Two per specification.** Each names the exact trigger and states exactly what the system returns.

Flowchart

Pick one of your five domains and draw a flowchart in Excalidraw (excalidraw.com). Export it as an image and include it in your PDF. Your flowchart must:

- Have a minimum of 5 steps
- Use standard shapes: rectangles for processes, diamonds for decisions, ovals for start/end
- Include at least one decision diamond with both exits clearly labelled (Yes / No)
- Have an unambiguous flow from start to end

How it is marked

Criterion	Marks	What we are looking for
All 5 specs with all 5 fields	**20**	Nothing missing. Every field filled.
Outputs specific enough to test	**25**	Word counts, named formats, measurable terms. No vague language.
Edge cases: exact trigger and message	**20**	Two per spec. Exact trigger, exact return message. Not ‘handle invalid input’.
Flowchart: min 5 steps, at least one decision	**20**	Standard shapes. Diamond with both exits labelled. Minimum 5 steps.
Constraints are specific throughout	**15**	Specific prohibitions, not general guidelines.
Total	**100**	**Weighted to 10% of the semester grade.**

Common mistakes to avoid

- Output described as ‘a helpful response’, which is not specific enough to test.
- An edge case that says ‘handle invalid input’ without an exact trigger and exact return message.
- A flowchart with fewer than 5 steps, no decision diamond, or unlabelled exits.
- Five specs that are near-copies of each other with only the domain name changed.
- Specs that describe what a human would do, not what an AI system should do.
- Constraints written as vague guidelines rather than specific prohibitions.

Submission template

Fill in, then export the whole document as a single PDF

Complete all five specifications below. On the final page, insert your Excalidraw flowchart image. Submit the whole document as a single PDF via the LMS.

STUDENT NAME	STUDENT ID
BATCH / SECTION	DATE SUBMITTED

Specification 1: Health	
Task	
Input	
Expected output	
Constraints	
Edge case 1	
Edge case 2	

Specification 2: Transport	
Task	
Input	
Expected output	
Constraints	
Edge case 1	
Edge case 2	

Specification 3: Education	
Task	
Input	
Expected output	
Constraints	
Edge case 1	
Edge case 2	

Specification 4: Food	
Task	
Input	
Expected output	
Constraints	
Edge case 1	
Edge case 2	

Specification 5: Scheduling	
Task	
Input	
Expected output	
Constraints	
Edge case 1	
Edge case 2	

Flowchart

Domain chosen	
Domain	

Flowchart image

Insert your exported Excalidraw image (PNG or SVG) here.

Checklist before you submit

- ☐ Minimum 5 steps (count every shape)
- ☐ At least one decision diamond present
- ☐ Both exits from every diamond labelled (Yes / No)
- ☐ Exported as an image (PNG or SVG) from Excalidraw and inserted on this page